- Visualize key insights such as the percentage of positive, neutral, and negative reviews.
- Highlight frequently mentioned keywords and topics associated with specific sentiments.

6. Deployment:
    - Develop a web-based or desktop application for sentiment analysis.
    - Allow users to input customer feedback or upload files for batch analysis.
    - Integrate APIs for real-time sentiment prediction.

7. Scalability and Optimization:
    - Use cloud services like AWS or Google Cloud for scalable data storage and model deployment.
    - Implement parallel processing or GPU acceleration for faster analysis of large datasets.

Outcome:

By implementing this project, businesses can gain actionable insights into customer sentiments, enabling them to improve products, services, and customer experiences. Sentiment analysis provides a scalable way to monitor brand reputation and address concerns proactively.

Example Code:

```
import pandas as pd
import numpy as np
import re
from sklearn.model_selection import train_test_split from sklearn.feature_extraction.text import
CountVectorizer from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report, accuracy_score # Step 1: Load Data
data = pd.read_csv('customer_reviews.csv') data['cleaned_reviews'] = data['review'].apply(lambda x:
re.sub(r'[^a-zA-Z\s]', '', x).lower()) # Step 2: Split Data
X_train, X_test, y_train, y_test = train_test_split(
    data['cleaned_reviews'], data['sentiment'], test_size=0.2, random_state=42
)
```